25900311

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

SCHEME-Discipline Specific Core Course (DSC)

MG3DSCCMA200 - AI TECHNIQUES FOR DATA ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions Answer all questions

Each question carries **2** marks

1. What is semi-structured data? Write an example. [K] / [CO1]

   Data that does not follow a strict schema but has some organisational structure using tags or markers. **Example:** JSON, XML.

2. Define Data Analysis. [K] / [CO1]

   The process of inspecting, cleaning, transforming, and modelling data to discover useful information and support decision-making.

3. State the difference between 1D and 2D array in Numpy. [K] / [CO2]

1D array	2D array
Single row of elements	Rows and columns (matrix form)
Example: np. array([1,2,3])	Example: np. array([[1,2,3],[4,5,6]])

4. What are the reserved keywords? Give examples. [K] / [CO2]

   Words with predefined meanings in Python that cannot be used as identifiers.

   Examples: if, for, while

5. Name the process of converting long-format data into wide-format data in Pandas. Give an example.

Process: Pivoting.

df.pivot(index='name',columns='sub.',values='mark') [U] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions: Answer any **5** questions

Each question carries **5** marks

6. List the secondary sources of data, describe how each can be collected, and discuss their potential limitations.

Sources:

- Government publications
- Company records
- Research journals
- Websites/databases

Collection: Accessing online repositories, libraries, archives, or reports.
Limitations: May be outdated, biased, or lack accuracy.

7. Describe some environmental impacts of AI and how can it be reduced [U] / [CO1]

Impacts: High energy consumption in data centres, carbon emissions, and electronic waste.

Reduction: Use green data centres, optimise algorithms, apply model compression, use Renewable energy

8. Explain the built-in list functions with examples. [U] / [CO2]

- len(list) – returns the number of elements.
- append(x) – adds item at the end.
- sort() – sorts the list in ascending order.

Example:

```
a=[3,1,2]
a.sort()
print(a)
```

9. Explain any three data analysis functions in Pandas with examples. [U] / [CO2]

- df.describe() – summary statistics.
- df.groupby() – group data by column.
- df.corr() – computes correlation.

Example: df.describe()

10. What is data interpretation in the data analysis life cycle? Why is the correct interpretation of analysis results important for organisations? [U] / [CO3]

The process of deriving meaningful insights from analysed data.
Importance: Helps organisations make correct decisions, avoid misjudgments, and ensure accurate strategies.

11. Why is data summarization important in data analysis, and how can it be performed? [U] / [CO3]

Condensing large datasets to highlight key information.
Importance: Simplifies analysis, aids comparison, supports decision-making.
Methods: Aggregation, descriptive statistics (mean(), sum(), describe()).

12. What is missing data? List any two reasons why missing data may occur. Give one example. [K] / [CO3]

Information not available for some variables or records.
Reasons:

- Human error or omission.
- Data corruption or transmission loss.

Example: Age field left blank in a survey.

[5x5 = 25]

Part C

Essay Questions Answer any **1** questions

Each question carries **15** marks

13. Explain if, if-else, nested if, if –elif ladder statements in Python with examples. [U] / [CO2]

if statement: Executes a block if the condition is true

```
x = 10
if x > 5:
  print("x is greater than 5")
```

if–else statement: Executes one block if the condition true, another if false

```
if x % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Nested if: One if inside another.

```
if x > 0:
    if x < 10:
        print("x is between 1 and 9")
```

if–elif ladder: Multiple conditions checked sequentially.

```
if x < 0:
    print("Negative")
elif x == 0:
    print("Zero")
else:
    print("Positive")
```

14.Illustrate with example how to create a grouped bar plot in Matplotlib with a suitable data set and interpret the plot. [U] / [CO3]

```
import matplotlib.pyplot as plt
import numpy as np
students = ['A', 'B', 'C']
math = [80, 70, 90]
science = [85, 75, 95]
x = np.arange(len(students))
width = 0.35
plt.bar(x - width/2, math, width, label='Maths')
plt.bar(x + width/2, science, width, label='Science')
```

```
plt.xlabel('Students')
plt.ylabel('Marks')
plt.legend()
plt.show()
```

The plot visually compares each student's performance across two subjects Maths and Science by displaying their marks side by side for easy comparison.

[15x1 = 15]